Abstract

India records 1.68 lakh road accident deaths annually — one life lost every four minutes — making it the country with the highest road fatality rate globally. Critically, 50% of these deaths are preventable during the 'Golden Hour' (the first 60 minutes post-trauma), yet average ambulance response times of 35–90 minutes render this window largely unused. Bystanders, already present at accident scenes, hesitate to help due to fear of legal consequences, despite India's Good Samaritan Law (2019) — a law 94% of Indians remain unaware of.

ResQ Network is an AI-powered emergency response platform that bridges this gap: automatic crash detection via smartphone sensors, GPS-based dispatch to the nearest trained bystander within seconds, multilingual AI voice guidance for on-scene first aid, and blockchain-based legal protection for responders. Hospitals are pre-notified with victim data before ambulance arrival, saving 10–15 minutes.

This report presents the complete Software Engineering documentation — feasibility analysis, requirements engineering (IEEE IEENSVR), system design with all UML and data diagrams, construction and testing strategy, risk analysis, SQA, configuration management, and project metrics — covering the full SDLC for ResQ Network.

Keywords: Emergency Response, Golden Hour, AI Dispatch, Good Samaritan Law, Bystander Activation, Blockchain, Mesh Network.

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AR	Augmented Reality
AES	Advanced Encryption Standard
CI/CD	Continuous Integration / Continuous Deployment
CPR	Cardiopulmonary Resuscitation
DFD	Data Flow Diagram
DPDP	Digital Personal Data Protection Act
ER	Entity-Relationship
ETA	Estimated Time of Arrival
FR	Functional Requirement
GDP	Gross Domestic Product
GPS	Global Positioning System
GSL	Good Samaritan Law
IEEE	Institute of Electrical and Electronics Engineers
IMU	Inertial Measurement Unit
LOC	Lines of Code
MoRTH	Ministry of Road Transport and Highways
NHAI	National Highways Authority of India
NFR	Non-Functional Requirement
OWASP	Open Web Application Security Project
P2P	Peer-to-Peer
SDLC	Software Development Life Cycle
SLA	Service Level Agreement
SQA	Software Quality Assurance
SRS	Software Requirements Specification
TDD	Test-Driven Development
TLS	Transport Layer Security
UML	Unified Modeling Language
VSL	Value of Statistical Life

Abbreviation	Full Form
WCAG	Web Content Accessibility Guidelines
YOLO	You Only Look Once (object detection)

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Chapter 1: Introduction

1.1 Overview of the Project

ResQ Network is an AI-powered emergency response platform designed to address India's road accident fatality crisis. India records approximately 1.68 lakh road accident deaths every year (MoRTH 2022), the highest road fatality rate globally — one death every four minutes.

The core insight: a trained bystander already at the accident scene can provide life-saving first aid far faster than any ambulance. ResQ Network harnesses this potential by combining AI, GPS dispatch, multilingual voice guidance, and legal protection to create the world's first proactive bystander-first emergency response system for India.

1.2 Problem Background

- Bystander Hesitation: 78% want to help but fear legal trouble; 94% unaware of Good Samaritan Law (2019).
- Slow Ambulance Response: 35–90 minute average ETA. Golden Hour (first 60 minutes) is routinely lost.
- Result: 71% of accident victims die before reaching hospital — all preventable deaths.

1.3 Objectives

1. Detect accidents automatically via smartphone/wearable sensors.
2. Dispatch the nearest trained ResQ Responder within seconds via GPS-based AI dispatch.
3. Guide the responder on-scene in their local language via AI voice + AR.
4. Pre-notify the hospital with victim data before ambulance arrival.
5. Provide blockchain-based legal protection to every responder.

1.4 Scope

- Flutter mobile app (victim + responder + hospital + admin modes).
- AI backend for crash detection, scene assessment, dispatch, and hotspot prediction.
- P2P mesh network for zero-signal highway operation.
- Integration with 112 India emergency services and hospital systems (HL7 FHIR).
- Blockchain layer (Hyperledger Fabric) for tamper-proof action logs.

1.5 Report Organisation

Chapter	Title
1	Introduction
2	Feasibility Study
3	Software Requirements Specification
4	System Design (with all diagrams)
5	SDLC Process

Chapter	Title
6	Risk Analysis
7	Software Quality Assurance
8	Configuration Management
9	Project Metrics
10	Expected Outcomes and Impact
11	Conclusion
Appendices A–C	SRS Sample, Test Cases, References

Chapter 2: Feasibility Study

2.1 Technical Feasibility

2.1.1 Technology Stack Availability

Component	Technology	Maturity
Mobile App	Flutter (Google)	Production-ready
Backend	Firebase Cloud Functions	Production-ready
AI Scene Analysis	YOLOv8 + TensorFlow Lite	Production-ready
Voice AI	Google Speech API	Production-ready
Blockchain	Hyperledger Fabric	Enterprise-grade
SMS / Call Fallback	Twilio	Production-ready
Mesh Network	Meshtastic Protocol (LoRa)	Active open-source
AR Training	ARCore / ARKit	Production-ready
Analytics	Google BigQuery	Production-ready
Hospital Integration	HL7 FHIR R4	International standard

2.1.2 Technical Challenges and Mitigations

Challenge	Mitigation
Zero-signal highway areas	Meshtastic P2P mesh network deployed at NHAI rest stops
AI crash detection false positives	Adaptive threshold + user confirmation; < 5% target false positive rate
Wearable sensor diversity	Phase 1: smartphone IMU only; Phase 2: wearable SDK integration
Hospital API heterogeneity	HL7 FHIR R4 standard adopted for all hospital integrations
Blockchain latency under high load	Hyperledger Fabric horizontal scaling; async logging

2.2 Operational Feasibility

2.2.1 Non-Functional Requirements Summary

Category	Requirement	Target
Performance	Dispatch alert delivery	< 5 seconds
Availability	System uptime	≥ 99.9%
Scalability	Concurrent emergencies	10,000+

Category	Requirement	Target
Reliability	False positive crash rate	< 5%
Security	Data encryption	AES-256 + TLS 1.3
Usability	Responder onboarding time	< 10 minutes
Localization	Voice guidance languages	10+ Indian languages
Portability	Platform support	Android 8.0+ / iOS 13+

2.3 Economic Feasibility (Cost Estimation)

Cost Category	Estimated Cost (INR)	Notes
Mobile App Development (Flutter)	₹8 – 12 Lakh	2 developers × 6 months
AI / ML Model Development	₹5 – 8 Lakh	YOLOv8 training + TFLite optimization
Backend & Cloud (Firebase + GCP)	₹3 – 5 Lakh / yr	Usage-based cloud billing
Blockchain Setup (Hyperledger)	₹4 – 6 Lakh	Node setup + SDK integration
AR Training Module	₹3 – 5 Lakh	ARCore / ARKit content
API Integrations (Twilio, Google)	₹1 – 2 Lakh / yr	Usage-based
Testing & QA	₹2 – 3 Lakh	Manual + automated
Legal & Compliance	₹1 – 2 Lakh	DPDP + IT Act advisory
Team & Operations (Year 1)	₹10 – 15 Lakh	Salaries, office, misc
Total (Pilot Phase)	₹40 – 60 Lakh	5-city pilot Year 1

Phase	Investment	Revenue / Funding Source
Pilot — Year 1	₹40 – 60 Lakh	CSR Grants + Angel Investors
Scale — Year 2 (5 Cities)	₹1.5 – 2 Crore	MoRTH Contract + Insurance Partners
National — Year 3 (25 Cities)	₹5 – 8 Crore	Series A + Government Licensing
International — Year 3+	₹15 – 20 Crore	Global licensing + WHO partnerships

2.4 Schedule Feasibility

2.4.1 Triple Constraint: Time, Cost, Quality

Parameter	Target	Strategy
Time	MVP in 6 months; full pilot in 12 months	Agile 2-week sprints; incremental feature delivery

Parameter	Target	Strategy
Cost	Within ₹40–60 Lakh for pilot	Open-source tools; cloud-native; lean 6-person team
Quality	< 5% false positives; > 99.9% uptime	TDD; staged rollout; continuous monitoring

2.4.2 Implementation Roadmap

Phase	Period	Key Deliverables
Q1–Q2 Pilot (Bangalore)	Months 1–6	Core App, ResQ Voice, ResQ Connect, 112 Integration
Q3–Q4 AI & Legal Layer (5 Cities)	Months 7–12	ResQ Vision, ResQ Shield, ResQ Learn, 50K Responders
Year 2 National Scale (25 Cities)	Months 13–24	ResQ Pulse, ResQ Grid, ResQ Predict, Government MoU
Year 3+ International	Months 25–36+	Indonesia, Bangladesh, Nigeria, Global ResQ Standard

Chapter 3: Software Requirements Specification (SRS)

3.1 Functional Requirements

FR ID	Requirement	Priority	Source
FR-01	Auto crash detection via smartphone accelerometer/gyroscope threshold breach	Critical	Victims
FR-02	Manual one-tap SOS trigger by bystander	Critical	Bystanders
FR-03	SOS relay via P2P mesh network when cellular signal absent	High	Highways
FR-04	GPS coordinates transmitted within 2 seconds of crash detection	Critical	Dispatch Engine
FR-05	AI scene photo analysis returns severity score within 3 seconds	Critical	Dispatch Engine
FR-06	Medical vault auto-unlocks on CRITICAL severity score	Critical	Hospital / Responder
FR-07	Nearest 3 trained ResQ Responders identified and pinged via GPS dispatch	Critical	Responder Network
FR-08	AI voice guidance delivered in responder's registered local language	Critical	Bystanders
FR-09	Step-by-step AR first-aid overlay on responder's camera view	High	Bystanders
FR-10	Ambulance dispatched; hospital pre-notified with victim data	Critical	Hospitals
FR-11	Every responder action timestamped and logged to Hyperledger Fabric	Critical	Legal Protection
FR-12	Downloadable blockchain legal certificate generated per assignment	High	Responders
FR-13	Real-time location and status updates sent to family contacts	High	Families
FR-14	Gamified AR-based CPR and first-aid training modules	Medium	Citizens
FR-15	AI-predicted accident hotspots pre-position responders	Medium	MoRTH / NHAI
FR-16	Integration with 112 India via bi-directional API for ambulance dispatch	Critical	Emergency Services
FR-17	Responder registration, certification, and annual recertification	High	Admin
FR-18	Admin dashboard with real-time incident monitoring	High	Admin
FR-19	Auto-escalate to 112 if no responder accepts	Critical	Dispatch Engine

FR ID	Requirement	Priority	Source
	within 60 seconds		
FR-20	Victim medical vault registration (blood group, allergies, medications, contacts)	High	Victims

3.2 Non-Functional Requirements

NFR ID	Category	Requirement	Target
NFR-01	Performance	Dispatch alert delivery	< 5 seconds from crash detection
NFR-02	Performance	AI severity scoring per photo	< 3 seconds
NFR-03	Availability	System uptime	≥ 99.9% (24 × 7)
NFR-04	Scalability	Concurrent active emergencies	≥ 10,000
NFR-05	Reliability	Crash detection false positive rate	< 5%
NFR-06	Security	Data at rest	AES-256 encryption
NFR-07	Security	Data in transit	TLS 1.3
NFR-08	Usability	New responder onboarding	< 10 minutes
NFR-09	Localization	Voice guidance languages	≥ 10 Indian languages
NFR-10	Portability	Platform support	Android 8.0+ and iOS 13+
NFR-11	Privacy	Compliance	Indian IT Act 2000 + DPDP Act 2023
NFR-12	Maintainability	P1 bug resolution SLA	< 24 hours
NFR-13	Testability	Unit test coverage (critical paths)	≥ 90%

Chapter 4: System Design

This chapter presents the complete visual system design of ResQ Network using standard software engineering diagrams. All diagrams follow UML 2.5 and DFD notation conventions.

4.1 System Architecture

ResQ Network is built on a four-layer architecture: Presentation (Flutter mobile + web), Application (Firebase Cloud Functions + AI Engine), Data (Firestore + Hyperledger Fabric), and Edge (Meshtastic mesh network for offline operation).

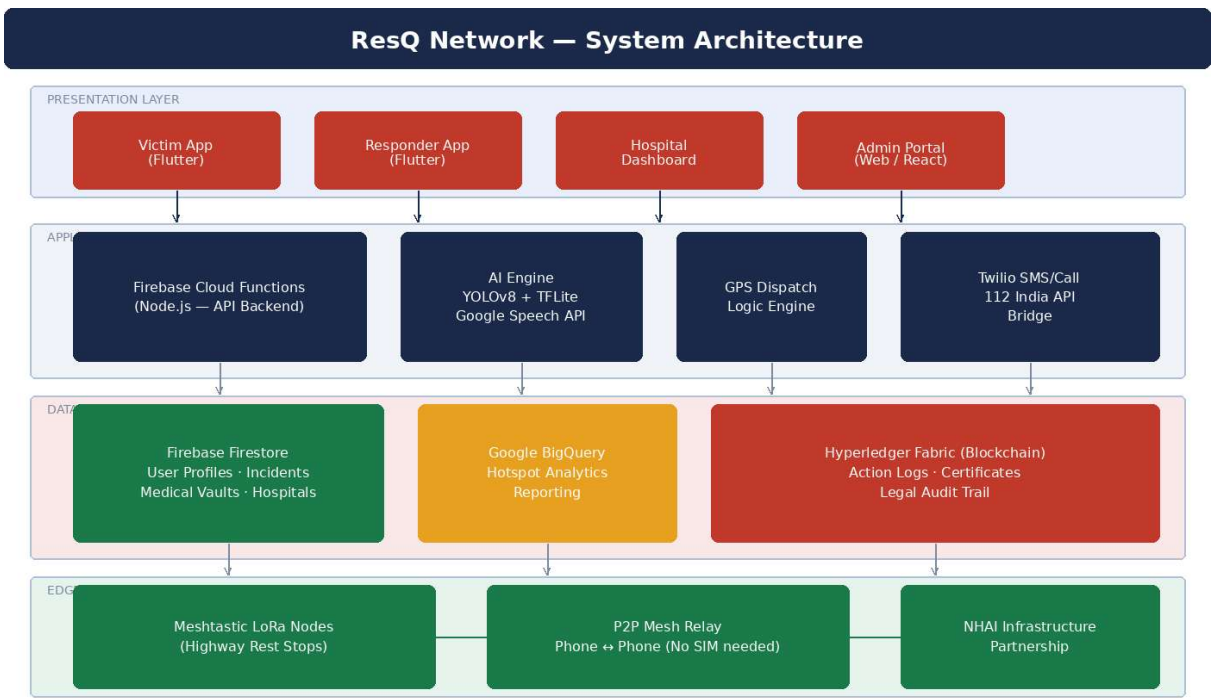


Figure 1: ResQ Network — System Architecture Diagram

4.2 Entity-Relationship (ER) Diagram

The ER diagram defines the 11 core data entities in the ResQ Network system, their attributes, and the relationships between them. Primary keys are shown in red; foreign keys in grey. Relationships are labelled with cardinality (1:1, 1:N, M:N).

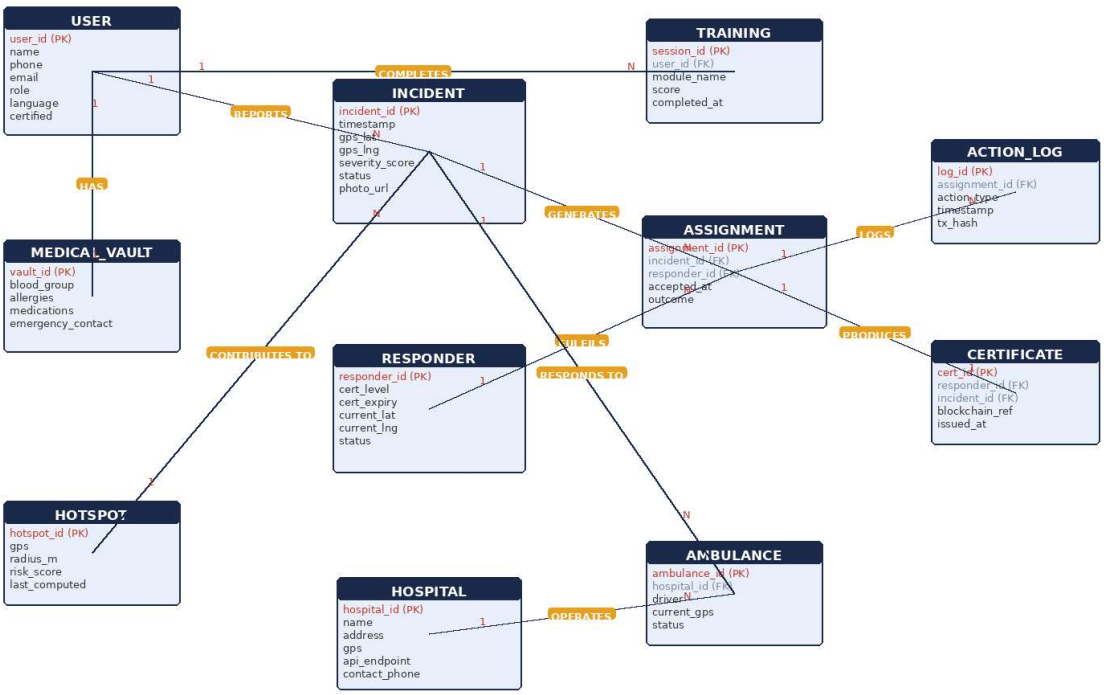


Figure 2: ResQ Network — Entity Relationship (ER) Diagram

Key Relationships:

- USER —— 1:1 —— MEDICAL_VAULT: Each registered user has exactly one medical vault.
- USER —— 1:N —— INCIDENT: A victim user can be involved in multiple incidents over time.
- INCIDENT —— M:N —— RESPONDER via ASSIGNMENT: Multiple responders can be dispatched to one incident.
- ASSIGNMENT —— 1:N —— ACTION_LOG: Every action taken during an assignment is logged individually.
- ASSIGNMENT —— 1:1 —— CERTIFICATE: One legal certificate is issued per completed assignment.
- HOSPITAL —— 1:N —— AMBULANCE: Each hospital manages multiple ambulances.

4.3 Data Flow Diagrams (DFD)

4.3.1 Level 0 — Context Diagram

The context diagram shows the entire ResQ Network system as a single process, with all external entities and the high-level data flows between them.

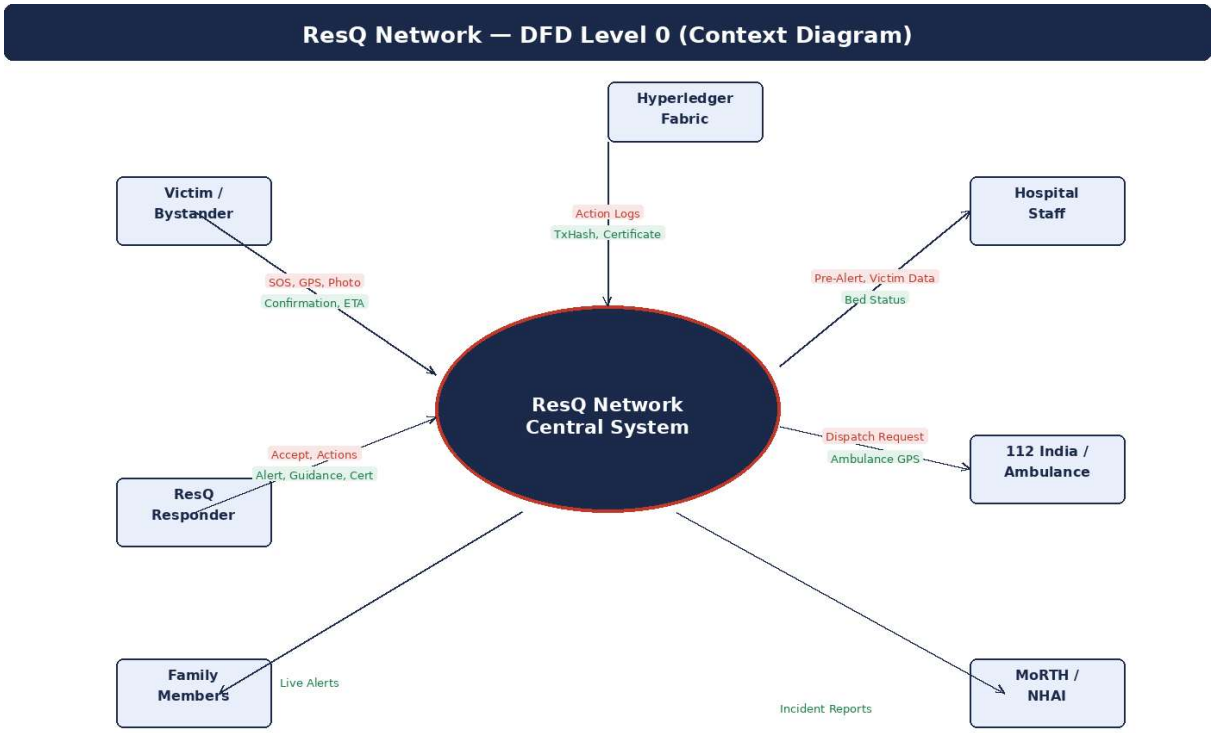


Figure 3: ResQ Network — DFD Level 0 (Context Diagram)

4.3.2 Level 1 — Major Processes

Level 1 DFD decomposes the system into six primary processing subsystems (P1–P6) and eight data stores (D1–D8), showing all data flows between them.

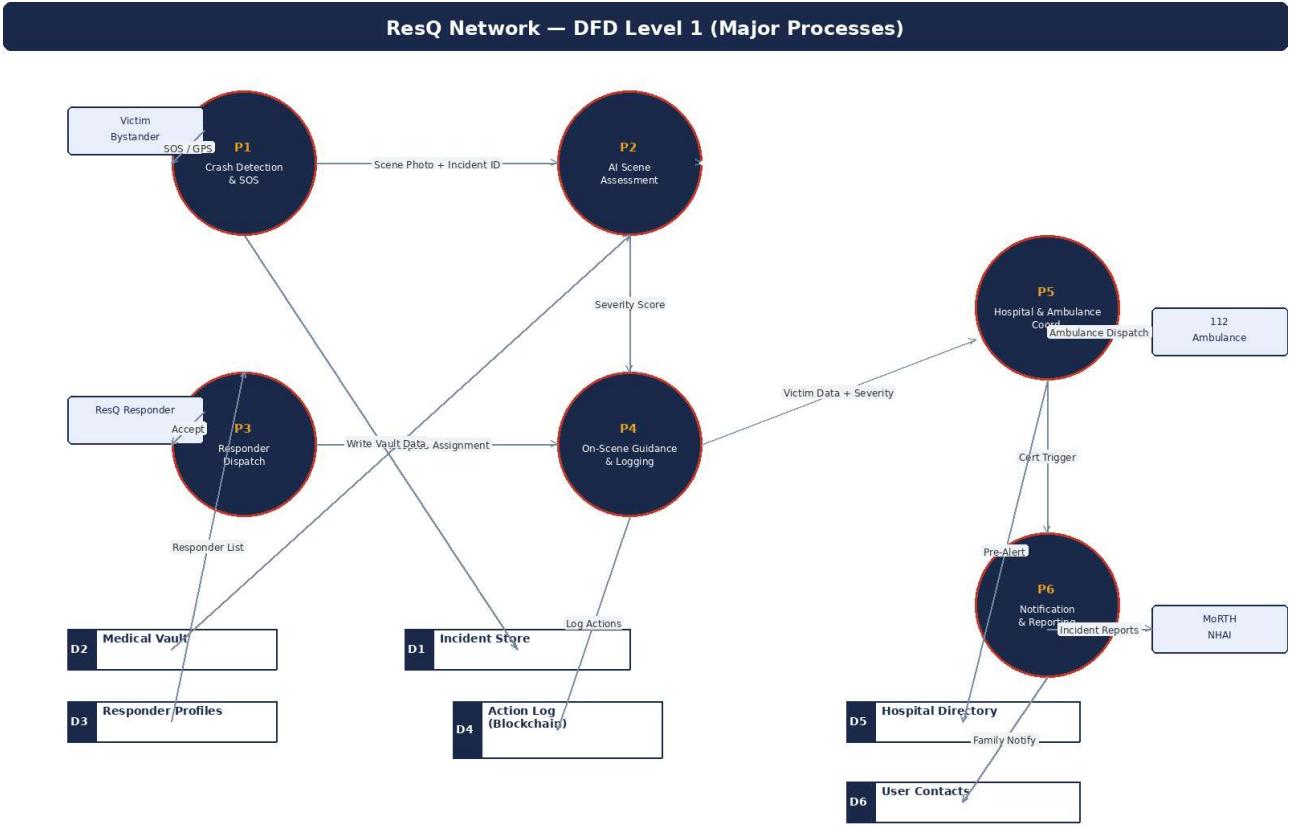


Figure 4: ResQ Network — DFD Level 1 (Major Processes and Data Stores)

Data Store	Name	Contents
D1	Incident Store (Firestore)	All incident records, status, GPS, photos, severity scores
D2	Medical Vault (Firestore — encrypted)	Blood group, allergies, medications, emergency contacts
D3	Responder Profiles (Firestore)	Location, certification level, availability status
D4	Action Log (Hyperledger Fabric)	Tamper-proof timestamped responder action records
D5	Hospital Directory (Firestore)	Hospital contact, GPS, API endpoint, trauma team capacity
D6	User Contacts (Firestore)	Registered emergency family contacts per user
D7	Certificate Store	Generated legal certificates with blockchain references
D8	Hotspot Store (BigQuery)	Computed accident hotspot polygons with risk scores

4.4 Use Case Diagram

The use case diagram models the interactions between five actor types and the fifteen major use cases of the

ResQ Network system. «include» relationships show mandatory sub-functions; «extend» relationships show optional extensions.

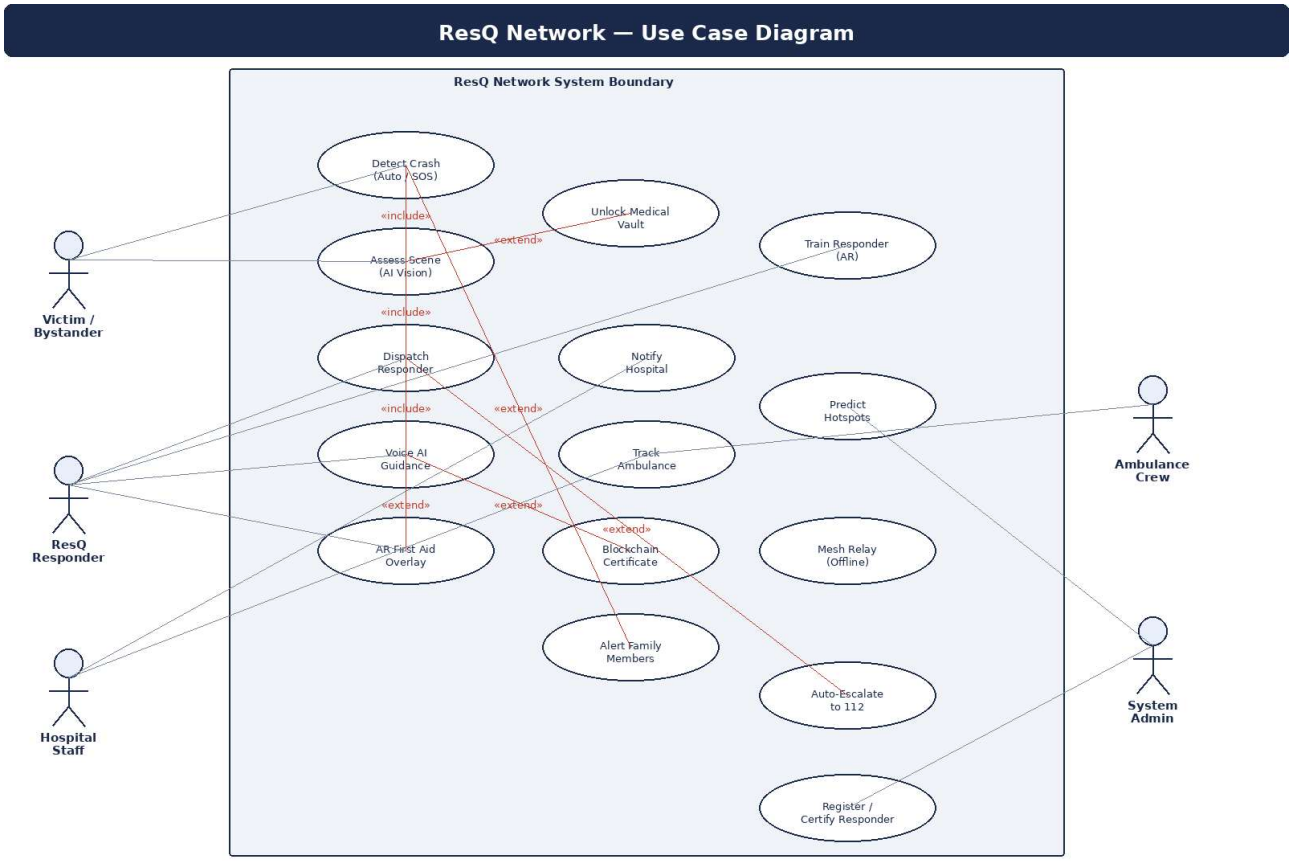


Figure 5: ResQ Network — Use Case Diagram

Use Case ID	Use Case Name	Primary Actor	Relationship
UC-01	Detect Crash (Auto / Manual SOS)	Victim / Sensor	Includes UC-02
UC-02	Assess Scene Severity (AI Vision)	System	Includes UC-03; Extends UC-05
UC-03	Dispatch Nearest Responder	System Dispatch Engine	Includes UC-04
UC-04	Guide Responder via Voice AI	ResQ Responder	Extends UC-08, UC-09
UC-05	Unlock Medical Vault	System (Auto)	Extends UC-02
UC-06	Notify Hospital in Advance	System	Includes UC-07
UC-07	Track Ambulance in Real-Time	Ambulance Crew	—

Use Case ID	Use Case Name	Primary Actor	Relationship
UC-08	Generate Blockchain Certificate	System (Hyperledger)	Extends UC-04 on completion
UC-09	AR First Aid Overlay	ResQ Responder	Extends UC-04
UC-10	Alert Family Members	System	Extends UC-01
UC-11	Train Responder (AR First Aid)	Citizen / Responder	—
UC-12	Predict Accident Hotspots	System AI	—
UC-13	Mesh Relay (Offline Mode)	System	Extends UC-01
UC-14	Auto-Escalate to 112	System	Extends UC-03 on timeout
UC-15	Register / Certify Responder	System Admin	—

4.5 Activity Diagram

The activity diagram traces the complete workflow of the primary accident response scenario from crash detection through patient handover, including all decision points and alternate flows.

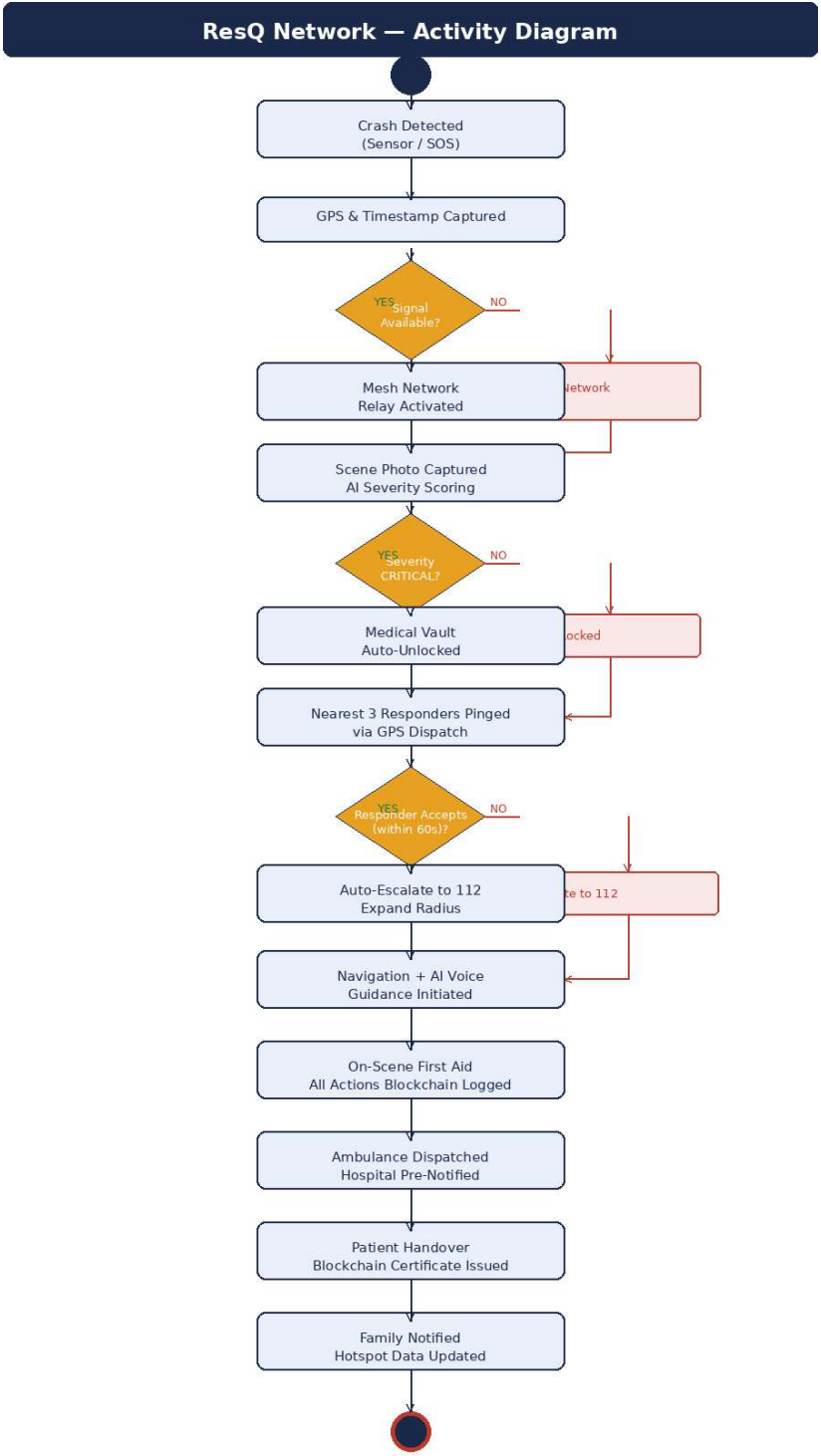


Figure 6: ResQ Network — Activity Diagram (Primary Accident Response Flow)

4.6 State Diagram — Incident Lifecycle

The state diagram models all possible states of an Incident object in the ResQ system, with the events that trigger transitions between states.

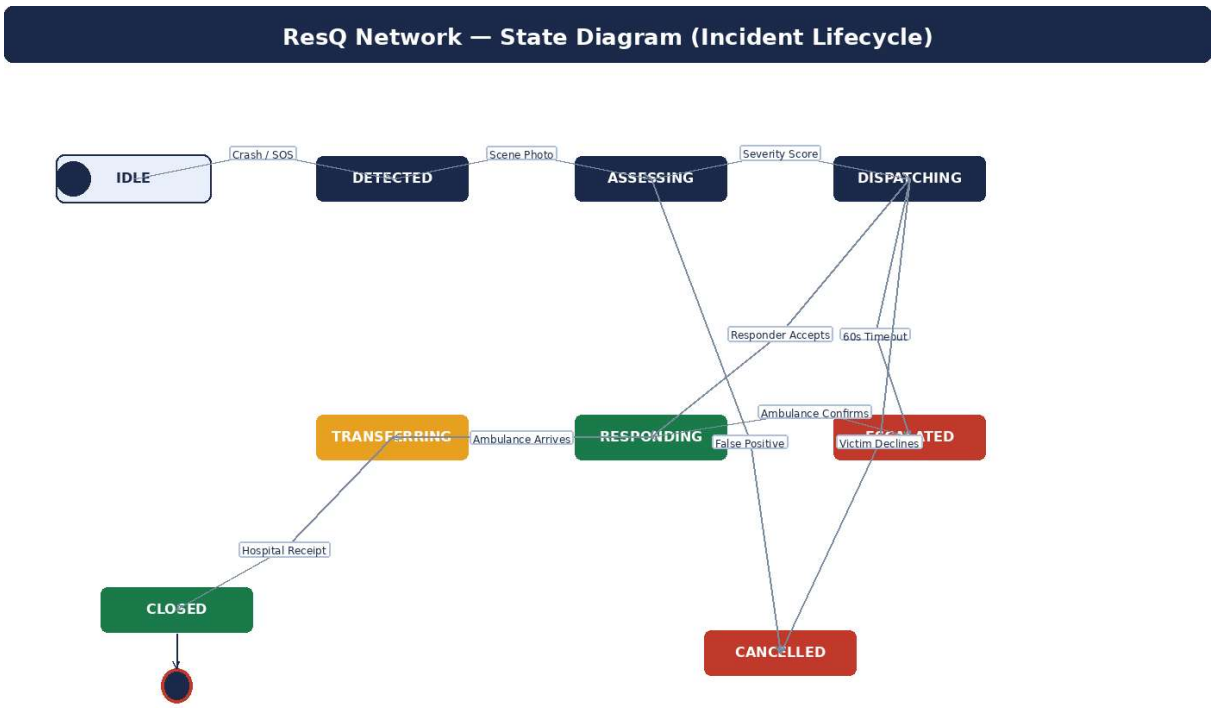


Figure 7: ResQ Network — State Diagram (Incident Lifecycle)

State	Description	Entry Event	Exit Event
IDLE	System monitoring; no active incident	System startup	Crash / SOS detected
DETECTED	Crash confirmed; location captured	Sensor / SOS trigger	Scene photo captured
ASSESSING	AI scoring in progress; medical vault queried	Photo uploaded	Severity score returned
DISPATCHING	Responders being alerted	Severity score computed	Responder accepts OR 60s timeout
ESCALATED	Auto-escalated to 112 India	60s timeout	Ambulance confirmed
RESPONDING	Responder on scene; guidance active	Responder GPS = scene GPS	Ambulance arrives

State	Description	Entry Event	Exit Event
TRANSFERRING	Patient handover to ambulance	Ambulance arrival confirmed	Hospital receipt confirmed
CLOSED	Incident complete; certificate issued	Hospital confirms receipt	—
CANCELLED	False positive or victim declines	Manual cancel / AI override	—

4.7 Sequence Diagram

The sequence diagram shows the time-ordered message flow between the seven principal system components for the primary accident response scenario.

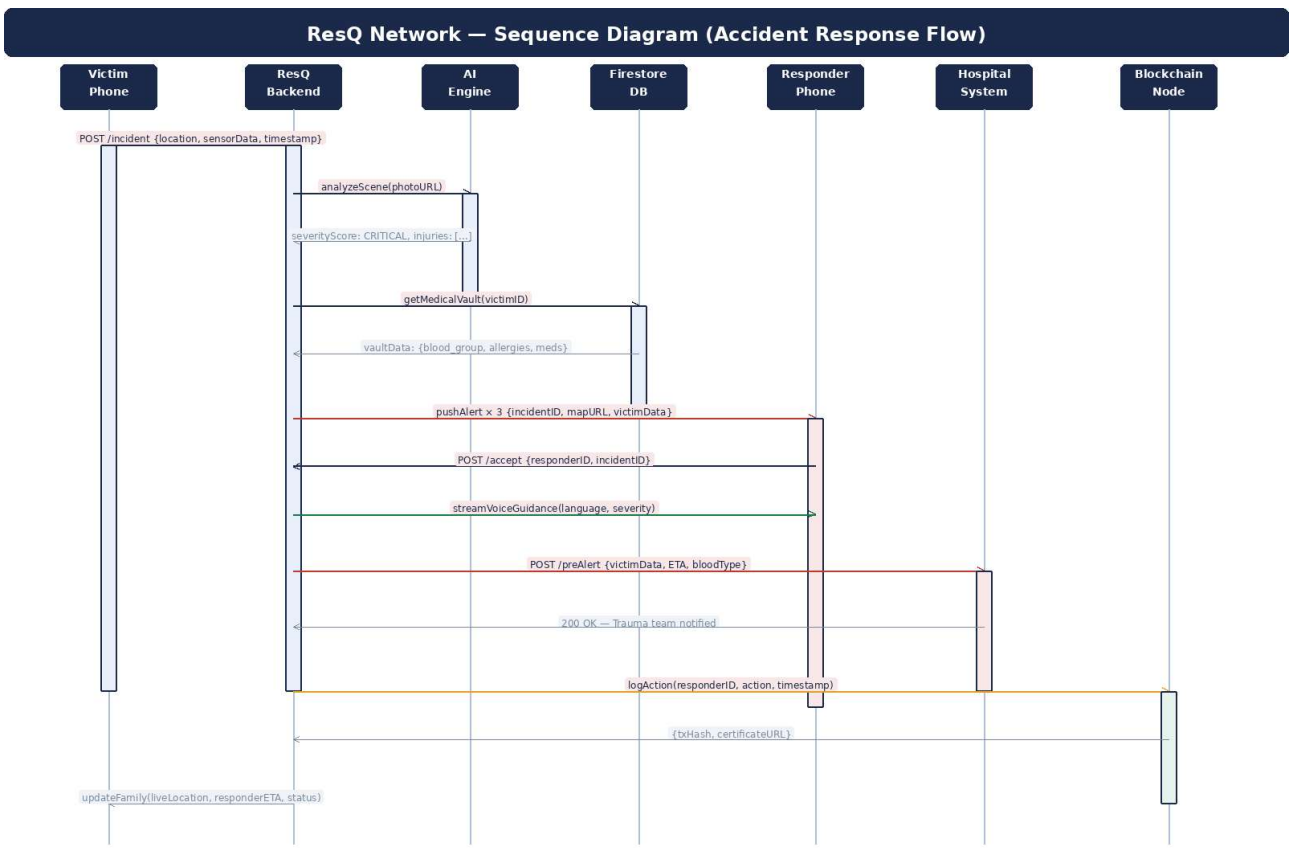


Figure 8: ResQ Network — Sequence Diagram (Accident Response Flow)

4.8 Class Diagram — Key Classes

Class	Key Attributes	Key Methods	Associates With
IncidentManager	incidentId, status, gps, severity	detectCrash(), assessScene(), closeIncident()	AIEngine, DispatchEngine, BlockchainLogger
AIEngine	modelVersion, threshold	analyzePhoto(): SeverityScore, detectCrash(): boolean	Called by IncidentManager
DispatchEngine	searchRadius, maxResponders	findNearestResponders(), pingResponder()	ResponderProfile, GpsService
ResponderProfile	location, certLevel, status	updateLocation(), acceptIncident()	User, Assignment
VoiceGuidanceService	language, severity	streamGuidance(): AudioStream	GoogleSpeechAPI
BlockchainLogger	fabricEndpoint	logAction(): TxHash, generateCertificate()	Hyperledger Fabric
MedicalVault	bloodGroup, allergies	unlock(incidentId), getEmergencyContact()	User, IncidentManager
MeshRelay	deviceId, peers	relay(SosPacket), broadcast(Alert)	NetworkService
HospitalNotifier	hospitalId, apiEndpoint	preAlert(), updateStatus()	IncidentManager
HotspotPredictor	model, dataSource	computeHotspots(), getHighRiskZones()	BigQueryService

4.9 User Interface Design

Screen	User	Key UI Elements	Design Principle
Home (Victim Mode)	Any user	Large SOS button, status indicator, hospital locator	One-tap emergency; WCAG 2.1 AA; high contrast
Responder Alert	ResQ Responder	Incident map, victim card, Accept/Decline, ETA	Urgent visual + haptic feedback; < 3 taps to accept
On-Scene Guidance	ResQ Responder	AI voice overlay, AR camera, action checklist	Hands-free capable; voice-first; large text
Hospital Dashboard	Hospital Staff	Incoming patient cards, blood type badge, ETA countdown	Real-time auto-refresh; colour-coded severity
Admin Portal	System Admin	Incident heatmap, responder network, KPIs	Data-dense; desktop-optimised; filter + export
Training Module	Citizen / Responder	AR CPR dummy, progress bar, badge collection	Gamified; rewarding; offline-capable

Chapter 5: SDLC Process

ResQ Network follows a Hybrid Agile-Waterfall SDLC — phased milestones with iterative 2-week sprints within each phase — appropriate for a safety-critical healthcare application.

5.1 Requirement Engineering (IEEE IEENSVR)

Step	Activity	Application in ResQ Network
I — Inception	Identify stakeholders	Victims, bystanders, hospitals, MoRTH, NHAI, insurance, NGOs, legal advisors
E — Elicitation	Gather requirements via surveys & interviews	Survey of 500+ bystanders; interviews with paramedics; observation of 112 India call center
E — Elaboration	Refine requirements	Use case modeling, scenario walkthroughs, low-fi prototyping
N — Negotiation	Resolve stakeholder conflicts	Hospital pre-notification scope; legal protection boundaries defined with legal advisors
S — Specification	Document SRS	SRS (Chapter 3): 20 FRs, 13 NFRs, 15 use cases
V — Validation	Verify completeness and feasibility	SRS reviewed by paramedics, legal counsel, AI engineers; scenarios traced to use cases
R — Risk Management	Identify and mitigate RE risks	Key risks: hospital API standards, GSL state variations, rural smartphone penetration

5.2 Design

Design decisions documented in Chapter 4. Key architectural decisions:

- Flutter for cross-platform reach with a single codebase.
- Firebase for real-time low-latency dispatch (sub-second pub/sub).
- Hyperledger Fabric (permissioned) over Ethereum (public) for legal-grade logging.
- Meshtastic LoRa protocol for highway mesh — proven in disaster relief contexts.
- HL7 FHIR R4 standard for all hospital API integrations.

5.3 Construction

5.3.1 Technology Stack

Layer	Technology	Purpose
Frontend	Flutter (Dart)	Cross-platform mobile app (Android + iOS)
State Management	Riverpod / BLoC	Reactive UI state management
Backend	Firebase Cloud Functions	Serverless API endpoints

Layer	Technology	Purpose
	(Node.js)	
Real-time DB	Firebase Realtime Database	Low-latency dispatch and tracking
Document DB	Firebase Firestore	User data, incidents, medical vaults
AI / ML	YOLOv8 + TensorFlow Lite	Scene analysis and crash detection
Voice AI	Google Speech-to-Text + TTS	Multilingual guidance
Blockchain	Hyperledger Fabric	Tamper-proof legal action logging
SMS / Calls	Twilio	Fallback notifications
Mesh Network	Meshtastic Protocol (LoRa)	Zero-signal emergency relay
AR Module	ARCore (Android) / ARKit (iOS)	First aid training
Analytics	Google BigQuery + Looker Studio	Hotspot prediction and reporting

5.4 Testing

Test Type	Scope	Tools	Pass Criteria
Unit Testing	Individual functions / classes	Flutter test; Jest	≥ 90% critical path coverage
Integration Testing	API interactions between services	Postman; Firebase Emulator	All API contracts satisfied
System Testing	End-to-end accident response	Staged simulation tests	Full scenario < 8 min
Performance Testing	Dispatch latency under load	JMeter; Firebase Performance	Alert < 5 sec at 10K concurrent users
Security Testing	Encryption; auth bypass	OWASP ZAP; manual pen test	No critical vulnerabilities
Usability Testing	Navigation; voice clarity	50+ user participants	≥ 85% task success rate
Regression Testing	Existing features post-update	GitHub Actions CI/CD	Zero regression failures
Field Testing	Real-world simulation (Bangalore)	Controlled mock scenarios	System behaves as designed
Localization Testing	Voice accuracy (10+ languages)	Native speaker validation	< 5% error rate per language
Blockchain Testing	Certificate generation & tamper-resistance	Hyperledger test network	Certs verifiable on ledger

5.5 Maintenance

Type	Description	ResQ Examples
Corrective	Fix post-deployment bugs	False positive fixes; dispatch patches; blockchain sync errors
Adaptive	Accommodate environment changes	New OS versions; 112 API changes; HL7 FHIR updates
Perfective	Enhance based on user feedback	New languages; improved AI; UI improvements
Preventive	Reduce future failures	Code refactoring; security audits; DB optimization

Chapter 6: Risk Analysis

Risks rated by Probability (1–5) and Impact (1–5). Risk Score = P × I. Scores ≥ 12 are High priority.

Risk ID	Description	Prob.	Impact	Score	Mitigation
R-01	High crash detection false positive rate causes alert fatigue	3	4	12	Adaptive threshold; user confirmation; ML on false positive feedback
R-02	Hospital API integration delays (non-standard systems)	4	4	16	HL7 FHIR standard; hospital-side SDK; dedicated integration support
R-03	Insufficient responder network in pilot cities	3	5	15	NSS/NCC/corporate partnerships; NGO collaborations; incentive programs
R-04	Government regulatory delays for 112 India API access	3	5	15	Early MoU negotiations; Twilio fallback; engage MoRTH as advisor
R-05	Blockchain latency spikes at high incident volumes	2	3	6	Horizontal scaling; async logging with eventual consistency
R-06	Medical vault data breach	2	5	10	AES-256; DPDP compliance audit; annual pen testing; zero-trust arch
R-07	Rural / highway areas with zero smartphone coverage	4	4	16	Meshtastic mesh at NHAI rest stops; NHAI partnership
R-08	User adoption slower than projected	3	4	12	Gamified onboarding; community health worker partnerships
R-09	Key team member attrition	2	4	8	Knowledge documentation; pair programming; competitive retention
R-10	AI model bias (night / rain conditions)	3	4	12	Diverse training data; night/rain augmentation; continuous retraining
R-11	Funding gap between pilot and scale phases	2	5	10	Multi-stream funding; insurance MoU; premium hospital API tier
R-12	Legal challenge to blockchain cert admissibility	2	4	8	Legal experts pre-launch; align with IT Act 2000 Section 85B

Chapter 7: Software Quality Assurance (SQA)

7.1 Quality Objectives

- Zero critical defects in dispatch and medical vault subsystems at launch.
- ≥ 90% unit test coverage on all safety-critical code paths.
- All NFRs verified through formal acceptance testing before each phase launch.
- OWASP Top 10 vulnerabilities addressed before production deployment.
- WCAG 2.1 AA accessibility compliance verified by external audit.

7.2 SQA Activities

Activity	Frequency	Responsible	Output
Requirements Review	Once per phase	Tech Lead + Domain Expert	Reviewed and signed-off SRS
Design Walkthrough	Once per phase	Full Team	Design review report
Code Review	Every pull request	2 peer reviewers	Approved / rejected PR with comments
Static Code Analysis	Every CI build	GitHub Actions (Dart Analyze, ESLint)	Analysis report; blocking on critical issues
Unit Test Execution	Every CI build	Automated	Test coverage report
Integration Test Execution	Every staging deployment	Automated (Postman / Firebase Emulator)	Integration test report
Security Audit	Quarterly	External pen tester + OWASP ZAP	Vulnerability report + remediation tracker
Performance Baseline	Before each phase launch	JMeter + Firebase Performance	Performance benchmark report
Usability Testing	End of each phase	UX Researcher + 50 participants	Usability report + priority issues list
Accessibility Audit	Before public release	External WCAG auditor	Accessibility compliance certificate

7.3 Quality Metrics

Metric	Definition	Target
Defect Density	Defects per 1,000 LOC	< 2 per KLOC at launch
Test Coverage	% of code covered by unit tests	≥ 90% (critical paths); ≥ 70% overall
P1 Resolution Time	Time from defect report to verified fix	< 24 hours
Mean Time Between Failures	Average time between system failures	> 720 hours (30 days)

Metric	Definition	Target
False Positive Rate	Crash alerts that are not actual crashes	< 5%
Dispatch Latency	Detection to responder notification	< 5 seconds (p95)
User Satisfaction	Post-interaction NPS survey	≥ 70 NPS
Blockchain Certificate Integrity	Certificates verifiable on ledger	100%

Chapter 8: Configuration Management

8.1 Version Control — Git Branching Strategy

Branch	Purpose	Protection Rules
main	Production-ready code	PR + 2 reviewers + all CI checks pass
staging	Pre-production testing	PR + 1 reviewer + integration tests pass
develop	Integration of completed features	PR + 1 reviewer + unit tests pass
feature/xxx	New feature development	Developer-owned; CI must pass before PR
hotfix/xxx	Critical production bug fixes	PR + Tech Lead approval; expedited review

8.2 Release Management

Release Type	Frequency	Versioning	Process
Major Release	Every 6 months (per roadmap phase)	v1.0, v2.0, v3.0	Full regression + UAT + staged rollout (5%→25%→100%)
Minor Release	Bi-monthly sprint completion	v1.1, v1.2 …	Integration tests + staging validation + 24hr soak
Patch / Hotfix	On-demand for P0 / P1 issues	v1.1.1, v1.1.2 …	Expedited review + targeted regression + immediate deploy
AI Model Update	Quarterly	model-v1.0, model-v2.0 …	A/B testing in production; rollback plan mandatory

8.3 CI/CD Pipeline

6. Developer pushes to feature branch → GitHub Actions triggers.
7. Static analysis (Dart Analyze, ESLint) — blocking on errors.
8. Unit tests executed — build fails if coverage drops below threshold.
9. Docker image built and pushed to Google Artifact Registry.
10. Deployed to Firebase staging environment.
11. Integration tests run against staging.
12. Manual QA sign-off required for staging → main promotion.
13. Production deployment via Firebase App Distribution (staged rollout).

Chapter 9: Project Metrics

9.1 LOC Estimation

Subsystem	Language	Estimated LOC	Complexity
Flutter Mobile App (Victim + Responder)	Dart	12,000 – 15,000	High
AI Scene Analysis Module	Python (YOLOv8 / TFLite)	3,000 – 5,000	Very High
Firebase Cloud Functions (API Backend)	Node.js / TypeScript	6,000 – 8,000	High
Blockchain Integration (Hyperledger)	Go / Node.js SDK	2,000 – 3,000	High
Voice Guidance Service	Node.js	1,500 – 2,000	Medium
AR Training Module	Dart + ARCore SDK	3,000 – 4,000	High
Mesh Network Relay	Dart + Meshtastic SDK	1,500 – 2,000	Medium
Hospital Notification Service	Node.js (HL7 FHIR)	2,000 – 2,500	Medium
Admin Dashboard (Web)	React + TypeScript	4,000 – 5,000	Medium
Total	—	35,000 – 46,500	—

9.2 Effort Estimation (COCOMO Basic)

- Estimated LOC: ~40 KLOC (mid-range)
- COCOMO Formula (semi-detached): Effort = 3.0 × (KLOC)^1.12
- Effort = 3.0 × (40)^1.12 ≈ 189 person-months
- Team of 6: Duration ≈ 31.5 months — aligns with 36-month roadmap
- Productivity: ≈ 211 LOC / person-month

9.3 Function Point Analysis

FP Category	Count	Weight	Weighted FP
External Inputs (user actions)	20	Avg (4)	80
External Outputs (alerts, certs, reports)	15	High (7)	105
External Queries (search, lookup)	12	Avg (4)	48
Internal Logical Files	8	High (15)	120
External Interface Files (APIs)	6	High (10)	60
Total Unadjusted FP	—	—	413
Complexity Adjustment (safety-critical)	—	× 1.15	—

FP Category	Count	Weight	Weighted FP
Adjusted Function Points (AFP)	—	—	≈ 475 AFP

Chapter 10: Expected Outcomes and Impact

10.1 Quantified Projections

Metric	Year 1 (5 Cities)	Year 3 (National)
Lives Saved	8,500+	30,000+ annually
Trained ResQ Responders	50,000	5,00,000+
Average Response Time	35 min → 4 min	< 3 min (urban)
Economic Value Created (VSL)	₹850 Crore	₹3,000+ Crore
Hospitals Pre-Notified	25+	500+
Legal Certificates Issued	8,500+	30,000+ annually

10.2 SDG Alignment

SDG	Goal	ResQ Network Contribution
SDG 3	Good Health and Well-Being	Saves lives through faster AI-guided emergency response
SDG 9	Industry, Innovation and Infrastructure	Builds India's first AI-guided emergency response infrastructure
SDG 10	Reduced Inequalities	Extends emergency response to all Indians including rural and highway areas
SDG 11	Sustainable Cities and Communities	Builds safer urban road ecosystems
SDG 17	Partnerships for the Goals	Government, private, and civil society partnerships for road safety

Chapter 11: Conclusion

ResQ Network is a convergence of technology, law, and human empathy — built to solve one of India's most preventable public health crises. This Software Engineering Report has documented every phase of the system's SDLC: feasibility, requirements, design with visual diagrams, construction, testing, risk analysis, SQA, configuration management, and project metrics.

The feasibility study confirms the system is technically achievable, operationally viable, economically justified, and schedulable within an 18-month pilot timeline. The requirement engineering process produced 20 FRs and 13 NFRs. Eight visual diagrams — architecture, ER, DFD L0 & L1, use case, activity, state, and sequence — provide a complete, implementation-ready design blueprint.

India has the law. India has the phones. India has the people. ResQ Network simply connects them — and in doing so, can save 30,000+ lives every year at national scale.

Appendix A: Sample SRS Entry (IEEE Std 830-1998)

FR-07: Dispatch Nearest ResQ Responder

Field	Detail
Requirement ID	FR-07
Title	Dispatch Nearest Trained ResQ Responder
Priority	Critical
Source	Accident Dispatch Engine / Responder Network
Description	Upon incident confirmation and AI scene assessment, the system shall identify the nearest 3 ResQ Responders within 2 km (configurable) and dispatch simultaneous push notifications. Each notification contains: incident map, victim severity, estimated time to scene, and visible medical alerts. First responder to accept is assigned; remaining alerts withdrawn.
Pre-conditions	Incident created (FR-01); GPS available; ≥ 1 AVAILABLE responder within search radius
Post-conditions	Assignment record created; responder status set BUSY; voice guidance stream initiated
Exception	No acceptance within 60 sec → FR-19 triggered; radius doubled and retried
Verification	Simulation: 10 staged incidents, 50 responders. PASS: dispatch < 5 sec in ≥ 95% of trials
Dependencies	FR-01, FR-02, FR-05, NFR-01

Appendix B: Sample Test Cases

TC ID	Test Name	Steps	Expected Result	Pass Criteria
TC-001	Auto Crash Detection — Positive	Simulate > 4G impact for > 200ms via test harness	SOS triggered; incident record created with GPS	PASS if incident created within 2 sec
TC-002	Crash Detection — False Positive	Simulate 2G bump < 50ms (pothole)	No SOS triggered; normal operation	PASS if no incident created
TC-003	Responder Dispatch Latency	Confirm incident; measure time to push notification on responder device	Notification received within 5 seconds	PASS if t ≤ 5000 ms at p95
TC-004	Medical Vault Auto-Unlock	Create CRITICAL incident; check responder view	Vault data visible (blood group, allergies)	PASS if vault data rendered on screen
TC-005	Blockchain Certificate Generation	Complete assignment; request certificate	Certificate generated with valid txHash	PASS if txHash resolves on Hyperledger ledger
TC-006	Offline Mesh Relay	No signal; two Meshtastic phones 500m apart; trigger SOS	Incident record created via mesh relay	PASS if record created within 30 sec
TC-007	Voice Guidance — Hindi	Set responder language to Hindi; accept incident	AI voice provides step-by-step in Hindi	PASS if native speaker confirms ≥ 95% accuracy
TC-008	Hospital Pre-Notification	Mark ambulance dispatched; check hospital dashboard	Patient card appears with blood type, ETA, severity	PASS if card appears within 10 sec

Appendix C: References

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